Task 1: Encrypt and decrypt a sample file using AES-256

You can use 7-Zip on Windows to create an encrypted archive.

Procedure
1. Create a text file named playlist.txt.
2. Add sample text, such as My favorite music playlists.
3. Install and open 7-Zip.
4. Right-click the file and choose 7-Zip → Add to archive.
5. Choose the archive format 7z.
6. Set a strong password.
7. Select AES-256 as the encryption method.
8. Enable Encrypt file names if available.
9. Create the archive and extract it using the correct password.
10. Compare the extracted text with the original file.

Task 2: Verify HTTPS using Wireshark

Use your own account and a browser. Do not attempt to capture another person's credentials.

Procedure
1. Open Wireshark and select your active network interface.
2. Start capturing packets.
3. Open a website you are authorized to access, such as your own account on a supported HTTPS website.
4. Enter the website's address beginning with https://.
5. Stop the capture.
6. Apply the display filter:
    tls
7. Inspect the packets for TLS traffic and the server connection.

You may also use:
    tcp.port == 443

Task 3: Scan a local web application using OWASP ZAP

Use a local application that you own or have permission to test.

Procedure
1. Install OWASP ZAP from its official website.
2. Start a local test application on your computer.
3. Open ZAP and select the automated scan option if suitable for the lab.
4. Enter your local application's URL, for example http://127.0.0.1:5000.
5. Run the scan against that local target.
6. Review the alerts and export the report.
7. Record the actual findings and their severity.

Possible findings in a deliberately vulnerable or misconfigured test application include:

Example issue	Potential impact	Recommended mitigation
Missing security headers	Reduced browser-side protection	Configure appropriate HTTP security headers.
Directory listing enabled	Unintended exposure of files	Disable directory indexing.
Missing cookie security flags	Increased cookie exposure risk	Configure Secure, HttpOnly, and suitable SameSite settings.

Task 4: Audit cloud storage sharing permissions

You can perform this audit in Google Drive or OneDrive.

Procedure
1. Sign in to your own account.
2. Review your files and folders.
3. Open the sharing or access-management panel for each relevant file.

4. Identify files shared publicly or using an “Anyone with the link” option.
5. Check whether each recipient needs continued access.
6. Remove unnecessary public links and change access to restricted or named users where appropriate.
7. Record your observations without including private document contents.

Sharing configuration	Risk	Recommended action
Anyone with the link	Link may be forwarded to unauthorized people	Restrict access to named users.
Publicly accessible document	Sensitive information may be exposed	Disable public access unless explicitly required.
Former employee or collaborator retains access	Unauthorized continued access	Remove obsolete permissions.

Task 5: Action plan for fixing directory listing
Issue: Directory listing is enabled on a web server, allowing visitors to view the names of files in a directory.

Action plan:
1. Confirm the finding in the vulnerability scan report.
2. Identify the affected web server or application route.
3. Disable directory indexing in the web server configuration.
4. Ensure that sensitive files are outside publicly accessible directories.
5. Restart or reload the service if required.
6. Rerun the scan and verify that directory listing is no longer reported.
7. Document the change and retain the before-and-after results.

For Apache, a common configuration change is to remove the Indexes option from the relevant directory configuration, for example:

```
<Directory "/var/www/html">
```

```
    Options -Indexes
</Directory>
```